

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 10)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

The Rio Bravo Conservation and Management Area in Belize exploits mahogany under a sustainable management plan while conserving a rainforest of over 200 tree, 400 bird and 70 mammal species.

0 1

. 1

Describe **five** features of the timber harvesting at Rio Bravo that make it sustainable.

[5 marks]

1

2

3

4

5

0 1

. 2

Explain why involving local communities is essential to the success of the reserve.

[3 marks]

0 1

. 3

Explain why selective logging is better for wildlife than clear-felling, and give **one** disadvantage.

[2 marks]

10

0 2

Table 1 shows characteristics introduced into commercial crops from crop wild relatives (CWRs).

Table 1

Crop	Wild relative source	Characteristic introduced
Sugar cane	Wild sugar cane, Indonesia	Resistance to mosaic virus
Tomato	Single wild species	Resistance to nine major diseases
Rice / barley	Wild salt-marsh varieties	Salt tolerance
Oil palm	Wild palms, central Africa	25% higher yield
Wheat	Wild wheat	Better mycorrhizal associations / phosphate uptake

0 2

. 1

Explain why commercial crop varieties often lack the characteristics shown in Table 1.

[3 marks]

0 2

. 2

Explain what is meant by a **Vavilov centre** (centre of diversity) and why these areas are conservation priorities.

[3 marks]

0 2 . 3

Explain why a large population does not necessarily have a large gene pool, and why a small gene pool is a problem.

[3 marks]

0 2 . 4

Give **one** wild plant species with potential as a new commercial food crop and the characteristic that makes it valuable.

[1 mark]

10

0 3

A conservation team is planning to protect a population of an endangered antelope in a reserve.

0 3 . 1

Explain why each of the following features of the population should be measured before a conservation strategy is decided.

[8 marks]

Population size

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Distribution

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Survival rate

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Age structure

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03 . 2 Describe how the population size of the antelope could be estimated.
[2 marks]

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10

04

Table 1 shows food conversion ratios (FCR) for livestock: the mass of feed needed to produce 1 kg of live mass gain.

Table 1

Livestock	FCR / kg feed per kg gain
Farmed salmon	1.2
Chicken	1.8
Pork	3.0
Beef (feedlot)	7.0
Sheep	8.0

04

1

Use Table 1 to explain why intensive poultry and fish production convert feed to food more efficiently than beef.

[3 marks]

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04

2

Explain **three** ways farmers reduce food chain energy losses in livestock production.

[3 marks]

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04

3

Explain why growing crops for direct human consumption usually produces more food per hectare than rearing livestock, and describe **one** situation where livestock is the better option.

[4 marks]

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10

0 5 . 1

Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.

[1 mark]

A Its mass

☐

B Its magnetic field

☐

C Its distance from the Sun

☐

D Its speed of rotation

☐

0 5 . 2

Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

[2 marks]

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0 5 . 3

Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

[2 marks]

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5

06 . 1

Explain why natural systems regulated by negative feedback rarely reach tipping points, and give an example of a natural tipping point in the geological past.

[3 marks]

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06 . 2

Explain why diverse systems are more resistant to change than simple systems, using a natural and a human example.

[4 marks]

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06 . 3

Give **three** positive feedback mechanisms by which human-induced warming may become self-sustaining.

[3 marks]

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0 7 . 1

Explain why carnivorous species such as salmon dominate aquaculture in MEDCs while herbivores such as carp and tilapia dominate in LEDCs.

[3 marks]

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0 7 . 2

Describe how all-female rainbow trout are produced and explain why this is done.

[3 marks]

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0 7 . 3

Explain how each of the following abiotic factors is controlled in an intensive salmon farm.

[4 marks]

Dissolved oxygen

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Water flow

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Day length

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Temperature

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10

08 . 1

Explain how each of the following agricultural changes affects wildlife habitats.
[6 marks]

Drainage of wet grassland

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Application of fertiliser to a species-rich meadow

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Removal of hedgerows to enlarge fields

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08 . 2

Explain why some farmland habitats are valuable for wildlife conservation.
[2 marks]

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08 . 3

Describe how agriculture contributes to climate change.

[2 marks]

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10

09 . 1

Describe the ecological features of mangroves and the adaptations of mangrove trees.

[3 marks]

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09 . 2

Explain **four** reasons why mangroves are important.

[4 marks]

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09 . 3

Describe the main threats to mangroves and the conservation efforts used.

[3 marks]

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10

10 . 1

Evaluate the extent to which the principles of the circular economy can be applied to **three** of the following: water supplies, mineral supplies, waste management, pollution control, energy supplies, food production.

[9 marks]

Do not write outside the box

Extra space

10 . **2** State what is meant by **industrial symbiosis**.

[1 mark]

Do not write outside the box

10

Write an essay on one of the following topics.

$$\boxed{1}\boxed{1}.\boxed{1}$$

Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

OR

1 1 . 2

Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

END OF QUESTIONS